2025-2026 – Econ 0016 – International Macroeconomics

Chapter 4: Terms of Trade, the World Interest Rate, Tariffs, and the Current Account

Schmitt-Grohé, Uribe and Woodford, "International Macroeconomics: A Modern Approach"

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Changes from version 1.0 are in red

0. Introduction

- We have studied the adjustment to endowment shocks.
- Other important shocks for open economies are shocks to
  - the terms of trade
  - the world interest rate
  - tariffs
- What happens with consumption, the trade balance, and the current account in response to these shocks?

- Thus far, we have studied an economy with a single good.
- Let's make the model more realistic and study an economy in which households like to consume a good different from the good they are endowed with.
- The relative price of exports in terms of imports is known as the terms of trade.
- For simplicity, assume that the consumption goods are *imported* and that the domestic endowments are *exported*.
- Note: not a good model to analyze Trump tariffs policy, as in this case imported goods are also produced domestically. We will consider production later in the course.

- Let P_1^X and P_1^M be the prices of exports and imports in period 1. The terms of trade in period 1, denoted TT_1, is

$$TT_1 \equiv \frac{P_1^X}{P_1^M}.$$

- As an example, the country exports "Mini" cars, imports tea and consumes only tea.
- The price of a Mini is £10,000 and the price of tea is £2000 per ton, then $P_1^X = 10,000$, $P_1^M = 2000$, and the terms of trade is 5, or $TT_1 = 10,000/2000 = 5$.
- Here, TT_1 represents the price of Minis in terms of tea.
- TT_1 indicates the amount of tea that the country can afford to import if it exports one Mini car.
- We assume that assets and liabilities (and therefore the NIIP B) are expressed in units of the imported good.

- The budget constraint in period 1 is
$$P_1^M C_1 + P_1^M B_1 - P_1^M B_0 = r_0 P_1^M B_0 + P_1^X Q_1.$$
and dividing by P_1^M
$$C_1 + B_1 - B_0 = r_0 B_0 + TT_1 Q_1.$$
- The budget constraint in period 2 is
$$C_2 + B_2 - B_1 = r_1 B_1 + TT_2 Q_2.$$
- Transversality condition:
$$B_2 = 0$$
- Intertemporal budget constraint
$$C_1 + \frac{C_2}{1+r_1} = (1+r_0)B_0 + TT_1 Q_1 + \frac{TT_2 Q_2}{1+r_1}$$
- It is identical to its counterpart in the one-good economy, except that here we have $TT_1 Q_1$ and $TT_2 Q_2$ instead of Q_1 and Q_2.

- Terms-of-trade shocks are just like output shocks.
- Interest rate parity holds so that $r_1 = r^\star$
- If income in period 1, $TT_1 Q_1$, goes up, the household doesn't care whether this is due to an increase in TT_1 or Q_1.
- Consequently, the adjustment to terms of trade shocks is identical to the adjustment to endowment shocks:
  - × The country finances temporary changes in the terms of trade by changing the current account (upwardly if the shock is positive and downwardly if it is negative), to smooth consumption over time.
  - × The country adjusts to permanent terms-of-trade shocks by mostly changing consumption (upwardly if the shock is positive and downwardly if it is negative), with little movement in the current account.

2. Terms-of-Trade Shocks and Imperfect Information

- When a shock hits the economy, it is not easy to tell whether it is permanent or temporary.
- Agents must form expectations about the duration of the shock, which may or may not be validated by future developments.
- When expectations are not fulfilled, the behavior of the economy may *ex-post* look at odds with the prediction of the intertemporal theory of current account determination.

2. Terms-of-Trade Shocks and Imperfect Information

- Suppose that initially $TT_1 = TT_2 = TT$ and $Q_1 = Q_2 = Q$.
- Suppose now that in period 1 the terms of trade appreciate by $\Delta > 0$ in period 1 and by $2 \times \Delta$ in period 2.
- How does the current account adjust to this development?
- If households expect the improvement in period 1 to be transitory ($TT_2 = TT$), then the current account in period 1 will improve, $CA_1 \nearrow$.
- But if households correctly anticipate that the future terms of trade will be even better by rising to $TT + 2 \times \Delta$ in period 2, then the current account in period 1 will deteriorate, $CA_1 \searrow$, as households will borrow against their higher expected future income.
- The takeaway from this hypothetical example is that what matters for the determination of the current account is the expected path of income, not just current income.
- This point is important for analyzing actual historical episodes, as the example on the next slides illustrates.

3. Imperfect Information, the Price of Copper and the Chilean CA

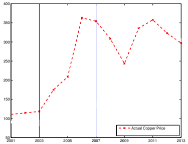
- Consider the dynamics of the Chilean current account during the copper price boom of the early 2000.
- We will show that the observed dynamics is consistent with the predictions of the intertemporal theory of the current account if one takes into account that Chile underestimated how long the copper price boom would last and to what heights the copper prices would rise.
- Copper is the main export product of Chile (more than 50% of exports). Thus, an increase in the world copper price represents a positive terms-of-trade shock for Chile.
- In the early 2000s, the price of copper began to rise (it more than tripled) after it had been relatively stable during the preceding two decades.

3. Imperfect Information, the Price of Copper and the Chilean CA

Actual Price of Copper, Chile, 2001–2013

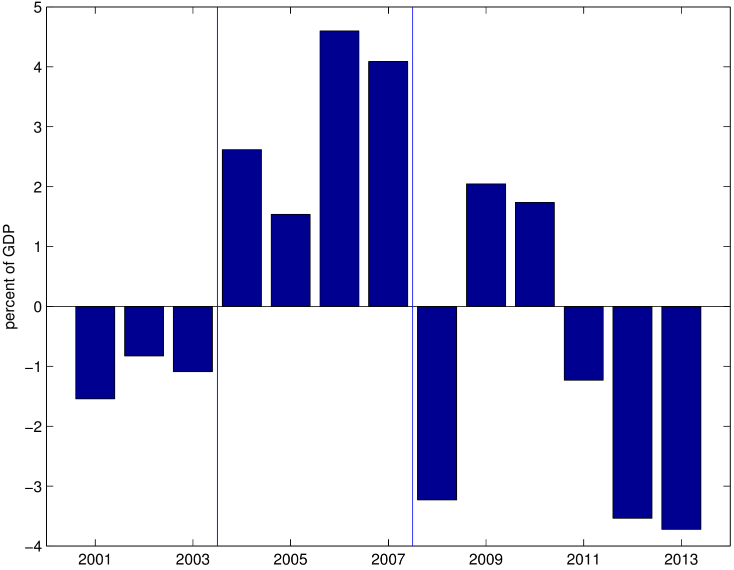
3. Imperfect Information, the Price of Copper and the Chilean Current Account

Forecast versus Actual Price of Copper, Chile, 2001–2013

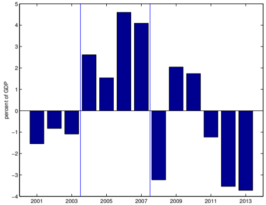


- Previous figure:
  - The crossed broken line shows the actual real price of copper in U.S. dollar cents of 2015 per metric pound (500 grams) from 2001 to 2013.
  - Between 2003 and 2007, the copper price rose from 120 to over 350 cents and then stayed that high until the end of the sample, 2013.
  - There was a dip during the global financial crisis of 2008, but it was temporary.
- Mapping this episode into our model
  - period 2003-2007 = period 1
  - years after 2007 = period 2
  - TT_1 increased and TT_2 also increased by as much as TT_1 if not more.
- Predictions of the intertemporal theory of the current account:
  - CA_1 and CA_2 should have deteriorated
  - What do we see in the data?

3. Imperfect Information, the Price of Copper and the Chilean CA
The Current Account, Chile, 2001-2013



3. Imperfect Information, the Price of Copper and the Chilean CA

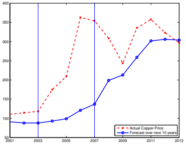


- We see that Chilean current account experienced a significant improvement between 2003 and 2007 from deficits of around one percent of GDP to surpluses of about three percent of GDP.
- This behavior of the CA contradicts, on the face of it, the predictions of the intertemporal theory of the current account.
- However, this prediction of the model hinges on the assumption that people had perfect foresight about the copper price.
- But the assumption that people could foresee that the rise of the price of copper would be long lasting turns out to be wrong.

3. Imperfect Information, the Price of Copper and the Chilean CA

Forecast versus Actual Price of Copper, Chile, 2001–2013

3. Imperfect Information, the Price of Copper and the Chilean CA



- How do we know?
- Figure on slide 14 plots the forecast of the average real price of copper over the next ten years produced by Chilean economic experts.
  - Until 2007 the experts expected the increase in the copper price to be transitory. For example, in 2005 experts predicted that the average copper price between 2005 and 2015 would be below 100.
  - Only by 2013, did forecasters seem to believe that high copper prices were there to stay.
- In light of these expectations about the path of the price of copper, the behavior of the current account is no longer in conflict with the predictions of the intertemporal model.
- The terms of trade were expected to improve only temporarily, therefore the current account should improve, which is what indeed happened.

4. World Interest Rate Shocks

- Movements in world interest rates have been identified as important factors driving business cycles and the external accounts in economies that are open to trade in goods and financial assets.
- Interest rate parity implies $r_1 = r^\star$
- What happens when the interest rate $r_1 = r^\star$ changes? Two potentially opposing effects:
  - × Substitution Effect
  - × Income Effect

$$C_1 + \frac{C_2}{1+r^\star} = (1+r_0)B_0 + TT_1 Q_1 + \frac{TT_2 Q_2}{1+r^\star}$$

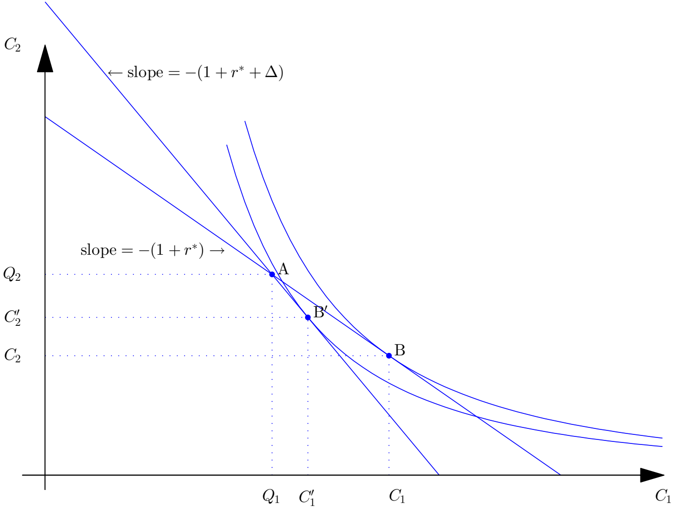
4. World Interest Rate Shocks

$$C_1 + \frac{C_2}{1+r^\star} = (1+r_0)B_0 + TT_1 Q_1 + \frac{TT_2 Q_2}{1+r^\star}$$

- Substitution Effect: A higher interest rate makes bonds more attractive and makes C_2 cheaper $\rightsquigarrow$ C_1 falls and saving increases.
- Income Effect:
  - × If households are borrowing in period 1, the interest rate hike makes them poorer (income effect is negative) $\rightsquigarrow$ consumption falls and saving increases. In this case, the income and substitution effects reinforce each other.
  - × If households are lending in period 1, the interest rate hike makes them richer (income effect is positive) $\rightsquigarrow$ consumption increases and saving falls. In this case, the income and substitution effects oppose each other.
- The net effect on C_1 depends on preferences, endowments and world interest rate.
- **Remark**: When the income and substitution effect oppose each other, we assume that preferences are such that the substitution effect dominates to ensure that saving is always an increasing function of the interest rate.

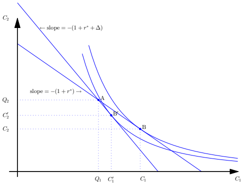
Adjustment to an Increase in the World Interest Rate

$(B_0 = 0)$



4. World Interest Rate Shocks

Adjustment to an Increase in the World Interest Rate



- See the figure of slide 18.
- Prior to the interest rate increase, the optimal consumption path is point B.
- Then the world interest rate increases from $r^\star$ to $r^\star + \Delta$.
- This causes the intertemporal budget constraint to rotate clockwise around the endowment point A (we are assuming that $B_0 = 0$).
- The old (before the increase in $r^\star$) optimal consumption path is point B.
- The new optimal consumption path is point B'.
- The increase in the interest rate causes period-1 consumption to fall from C_1 to C_1' and period-2 consumption to increase from C_2 to C_2'.
- Thus, the trade balance $TB_1' = Q_1 - C_1'$ and the current account $CA_1' = TB_1' + r_0 B_0$ both improve.

5. Import Tariffs

- Does an increase in import tariffs reduce imports and thereby improve the trade balance?
- It turns out that an increase in import tariffs can have either a positive, a negative, or no effect on a country's trade balance.
- What matters is not the import tariff per se but how it compares to expected future import tariffs.
- Intuitively, if the current import tariff is higher than the future one, then imports are relatively expensive in the current period and import demand should fall. This would improve the trade balance in the current period.
- But if the imposition of import tariffs is permanent, then there is no reason to shift consumption across time periods, and the import tariff will have no effect on the trade balance.

5. Import Tariffs

- How to embed an import tariff in our model?
- Assume that there are two goods, an export good and an import good as in the analysis of terms-of-trade shocks.
- Consumption is made of imported goods, while endowments are exported as they bring no utility.
- Assume that the government rebates the revenues generated by the import tariff to households in a lump sum fashion.
- Let τ_t denote the import tariff in period $t = 1, 2$.
- Let L_t denote the lump sum transfer in period $t = 1, 2$.

5. Import Tariffs

- Household budget constraints in periods 1 and 2:

$$(1+\tau_1)C_1 + B_1 = TT_1 Q_1 + L_1 + (1+r_0)B_0 \tag{1}$$

$$(1+\tau_2)C_2 + B_2 = TT_2 Q_2 + L_2 + (1+r_1)B_1 \tag{2}$$

- Transversality condition $B_2 = 0$.
- Intertemporal budget constraint

$$(1+\tau_1)C_1 + \frac{(1+\tau_2)C_2}{1+r_1} = \overline{Y} \tag{3}$$

where

$$\overline{Y} = (1+r_0)B_0 + TT_1 Q_1 + L_1 + \frac{(TT_2 Q_2 + L_2)}{1+r_1}$$

is lifetime wealth, which the household takes as given.
- The slope of the household intertemporal budget constraint is now $-\frac{(1+\tau_1)}{(1+\tau_2)}(1+r_1)$.

$$\max_{C_1, C_2} U(C_1) + \beta U(C_2)$$

subject to

$$(1+\tau_1)C_1 + \frac{(1+\tau_2)C_2}{1+r_1} = \overline{Y}. \tag{3}$$

- Solve (3) for C_2 and use resulting expression to replace C_2 from the household's lifetime utility function:

$$\max_{C_1} U(C_1) + \beta U\Bigg(\underbrace{\frac{1+r_1}{1+\tau_2}(\overline{Y} - (1+\tau_1)C_1)}_{C_2}\Bigg)$$

- The first-order condition associated with this problem is the Euler equation:

$$U'(C_1) = \frac{1+\tau_1}{1+\tau_2}\beta(1+r_1)U'(C_2). \tag{4}$$

5. Import Tariffs
Government

- The government rebates any revenue from the import tariff to households in a lump sum fashion
- Government budget constraint in period $t = 1, 2$:

$$L_t = \tau_t C_t$$

- Combining this expression with the households' intertemporal budget constraint

$$(1+\tau_1)C_1 + \frac{(1+\tau_2)C_2}{1+r_1} = (1+r_0)B_0 + TT_1 Q_1 + L_1 + \frac{(TT_2 Q_2 + L_2)}{1+r_1}$$

  gives the economy-wide resource constraint

$$C_1 + \frac{C_2}{1+r_1} = (1+r_0)B_0 + TT_1 Q_1 + \frac{(TT_2 Q_2)}{1+r_1}, \qquad (5)$$

  which is the same as in the economy without tariffs.
- This is intuitive because the government returns the tariff revenue to its citizens (who paid for the tariffs).

- An equilibrium is a consumption path (C_1, C_2) and a domestic interest rate r_1 such that the Euler equation (4) holds, the economy wide resource constraint (5) is satisfied and interest rate parity holds, that is,

$$U'(C_1) = \frac{1+\tau_1}{1+\tau_2}\beta(1+r_1)U'(C_2), \tag{4}$$

$$C_1 + \frac{C_2}{1+r_1} = (1+r_0)B_0 + TT_1 Q_1 + \frac{TT_2 Q_2}{1+r_1}, \tag{5}$$

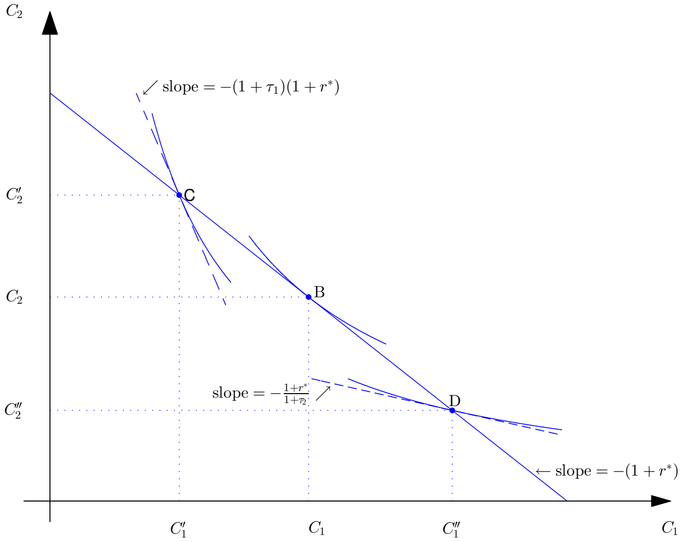
$$r_1 = r^\star \tag{6}$$

  given τ_1, τ_2, $TT_1 Q_1$, $TT_2 Q_2$, $(1+r_0)B_0$, and $r^\star$.
- We can now analyze the adjustment to changes in import tariffs.

- Initially $\tau_1 = \tau_2 = 0$.
- Then a temporary tariff is imposed: $\tau_1 > 0$ and $\tau_2 = 0$.
- Note that the slope of the household intertemporal budget constraint changes from $-(1 + r^\star)$ to $-(1 + \tau_1)(1 + r^\star)$.
- The economy was at point B on the figure of slide 27.
- It now moves to point C in the same figure

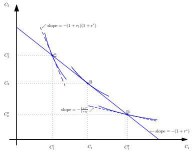
5. Import Tariffs

Adjustment to changes in import tariffs



5. Import Tariffs

A Temporary Increase in Import Tariffs



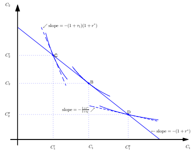
- Prior to the imposition of import tariffs ($\tau_1 = \tau_2 = 0$) the optimal consumption path is at point B.
- Changes in import tariffs leave the intertemporal resource constraint (the downward sloping solid line) unchanged.
- A temporary increase in import tariffs ($\tau_1 > 0$ and $\tau_2 = 0$) pushes the optimal consumption path to point C, where the slope of the indifference curve (the broken line) is steeper than at point B, $(1+\tau_1)(1+r^\star) > (1+r^\star)$.
- The increase in import tariffs causes period-1 consumption to decline ($C_1' < C_1$), and the trade balance in period 1 to improve ($TT_1Q_1 - C_1' > TT_1Q_1 - C_1$).
- The indifference curve associated with point C lies southwest of that associated with point B implying that the tariff is welfare reducing.

- Initially $\tau_1 = \tau_2 = 0$.
- Then a permanent tariff is imposed: $\tau_1 = \tau_2 > 0$.
- Note that the slope of the household intertemporal budget constraint does not changes: $-(1+\tau_1)/(1+\tau_2)(1+r^\star) = -(1+r^\star)$.
- The economy was at point B on the figure of slide 27.
- It stays at point B in the same figure
- The permanent increase in import tariffs has no effect on the equilibrium consumption path because the import tax rates cancel out of the Euler equation (4).

- Initially $\tau_1 = \tau_2 = 0$.
- Then a temporary tariff will be imposed in period 2: $\tau_1 = 0$, $\tau_2 > 0$.
- Note that the slope of the household intertemporal budget constraint changes from $-(1 + r^\star)$ to $(1 + r^\star)/(1 + \tau_2)$.
- The economy was at point B on the figure of slide 27.
- It moves to point D in the same figure

5. Import Tariffs

An Anticipated Future Increase in Import Tariffs



- Changes in import tariffs leave the intertemporal resource constraint (the downward sloping solid line) unchanged.
- An anticipated future increase in import tariffs ($\tau_2 > 0$ and $\tau_1 = 0$) pushes the optimal consumption path to point D, where the slope of the indifference curve is flatter, $(1 + r^\star)/(1 + \tau_2) < (1 + r^\star)$.
- The expected future increase in import tariffs causes period-1 consumption to increase from C_1 to C_2'' and the trade balance to deteriorate in period 1.
- The indifference curve associated with point D lies southwest of that associated with point B implying that the tariff is welfare reducing.

6. Summing Up

a. This chapter analyzes the effects of terms-of-trade shocks and interest-rate shocks in the context of the intertemporal model of the current account developed in chapter 3.

b. The terms of trade is the relative price of export goods in terms of import goods.

c. Terms-of-trade shocks have the same effects as endowment shocks: the economy uses the current account to smooth consumption over time in response to temporary terms of trade shocks, and adjusts consumption with little movement in the current account in response to permanent terms-of-trade shocks.

6. Summing Up

d. Interest rate shocks have a substitution and an income effect.
e. By the substitution effect, an increase in the interest rate discourages current consumption and incentivizes savings causing the current account and the trade balance to improve.
f. The income effect associated with an increase in the interest rate depends on whether households are borrowing or lending.
g. If households are borrowing, an increase in the interest rate has a negative income effect, as it makes borrowers poorer. As a result, consumption falls and the trade balance and the current account improve. In this case, the income and substitution effect go in the same direction.
h. If households are lending, the income effect associated with an increase in the interest rate is positive and leads to higher consumption and a deterioration in the trade balance and the current account. In this case, the income and substitution effects go in the opposite direction, partially offsetting each other.
i. Under log-preferences the substitution effect dominates (see problem set 2).

6. Summing Up

j. An increase in import tariffs need not improve the trade balance or the current account.
k. Only if import tariffs are temporary, will they lead to an improvement in the trade balance and the current account.
l. Future expected increases in import tariffs deteriorate the trade balance.
m. Permanent changes in tariffs leave the trade balance unchanged.
n. In the present framework, import tariffs are welfare decreasing.